MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics **CBSE CLASS 10 – WEEKLY CHAPTER TEST – 03** **Time:** 45 Minutes

Class: Class X (CBSE) **Chapter 4: Quadratic Equations** **MM: 20**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

Section A: MCQ & Assertion-Reason Questions

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Q1. If the roots of the equation $4x^2 - 2x + (k-4) = 0$ be reciprocal of each other, then k equals: [1]

(A) -8 **(B)** 8
(C) 4 **(D)** -4

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 8

Let the roots of $4x^2 - 2x + (k-4) = 0$ be α and $\frac{1}{\alpha}$ $(\alpha \neq 0)$.
Here $a = 4$, $b = -2$, and $c = k - 4$.

Formula Used: Product of roots $(\alpha \cdot \beta) = \frac{c}{a}$

Applying the product of roots relation:

$$\alpha \cdot \frac{1}{\alpha} = \frac{k-4}{4} \implies 1 = \frac{k-4}{4} \implies k - 4 = 4 \implies \boxed{k = 8}$$

Thus, option **(B)** is correct. **[1 mark]**

Q2. If the discriminant of the quadratic equation $3x^2 - 2x + c = 0$ is 16, then the value of c is: [1]

(A) 1 **(B)** -1
(C) 2 **(D)** -2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) -1

Comparing $3x^2 - 2x + c = 0$ with $ax^2 + bx + c_0 = 0$, we have $a = 3$, $b = -2$, and constant term c.

Formula Used: Discriminant $D = b^2 - 4ac$

Given $D = 16$:

$$(-2)^2 - 4(3)(c) = 16 \implies 4 - 12c = 16 \implies -12c = 12 \implies \boxed{c = -1}$$

Hence, option **(B)** is correct. **[1 mark]**

Q3. A quadratic equation whose one root is 2 and the sum of whose roots is zero, is: [1]

(A) $x^2 + 4 = 0$ **(B)** $x^2 - 4 = 0$

(C) $4x^2 - 1 = 0$ **(D)** $x^2 - 2 = 0$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B)** $x^2 - 4 = 0$

Let the roots be $\alpha = 2$ and β. Given $\alpha + \beta = 0 \implies 2 + \beta = 0 \implies \beta = -2$.

Then the product of roots is $\alpha\beta = (2)(-2) = -4$.

Formula Used: Required quadratic equation: $x^2 - (\alpha + \beta)x + \alpha\beta = 0$

Substituting $\alpha + \beta = 0$ and $\alpha\beta = -4$:

$$x^2 - (0)x + (-4) = 0 \implies \boxed{x^2 - 4 = 0}$$

Hence, option **(B)** is correct. **[1 mark]**

Q4. The quadratic equation $x^2 - ax + 1 = 0$ has two distinct real roots if: [1]

(A) $|a| = 2$ **(B)** $|a| < 2$

(C) $|a| > 2$ **(D)** None of these

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(C)** $|a| > 2$

Comparing $x^2 - ax + 1 = 0$ with $Ax^2 + Bx + C = 0$, we have $A = 1$, $B = -a$, and $C = 1$.

Formula Used: Two distinct real roots $\iff D = B^2 - 4AC > 0$

Setting $D > 0$:

$$(-a)^2 - 4(1)(1) > 0 \implies a^2 - 4 > 0 \implies a^2 > 4 \implies \boxed{|a| > 2}$$

(That is, $a > 2$ or $a < -2$.) Hence, option **(C)** is correct. **[1 mark]**

Q5. Assertion-Reason: [1]

Assertion (A): If one root of the quadratic equation $x^2 - 4x + 1 = 0$ is $2 + \sqrt{3}$, then the other root is $2 - \sqrt{3}$.

Reason (R): Irrational (surd) roots of a quadratic equation with rational coefficients always occur in conjugate pairs ($p + \sqrt{q}$ and $p - \sqrt{q}$).

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true, but R is not the correct explanation of A.

(C) A is true, R is false.

(D) A is false, R is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).**

For $x^2 - 4x + 1 = 0$, all coefficients $(1, -4, 1)$ are rational. Using the quadratic formula:

Formula Used: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(1)}}{2(1)} = \frac{4 \pm \sqrt{12}}{2} = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$$

Notice also that $(2 + \sqrt{3}) + (2 - \sqrt{3}) = 4 = -\frac{b}{a}$ and $(2 + \sqrt{3})(2 - \sqrt{3}) = 4 - 3 = 1 = \frac{c}{a}$.

Thus Assertion (A) is **true**, Reason (R) is **true** (explaining why irrational roots with rational coefficients appear as $\frac{-b \pm \sqrt{D}}{2a}$ conjugate pairs), and Reason (R) correctly explains Assertion (A). **[1 mark]**

Q6. Find the discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$, and hence find the nature of its roots. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Comparing $2x^2 - 4x + 3 = 0$ with standard form $ax^2 + bx + c = 0$, we get $a = 2$, $b = -4$, and $c = 3$.

Formula Used: Discriminant $D = b^2 - 4ac$

Substituting a, b, and c:

$$D = (-4)^2 - 4(2)(3) = 16 - 24 = -8 \quad \textbf{[1 mark]}$$

Since $D = -8 < 0$, the given quadratic equation has **no real roots**.

Therefore, $\boxed{D = -8 \text{ and the equation has no real roots}}$. **[1 mark]**

Q7. Using the quadratic formula, solve for x: $9x^2 - 9(a+b)x + (2a^2 + 5ab + 2b^2) = 0$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Comparing with $Ax^2 + Bx + C = 0$, we identify $A = 9$, $B = -9(a+b)$, and $C = 2a^2 + 5ab + 2b^2$.

Formula Used: Quadratic Formula: $x = \dfrac{-B \pm \sqrt{D}}{2A}$, $\quad D = B^2 - 4AC$

Step 1: Compute the discriminant D:

$$\begin{aligned} D &= [-9(a+b)]^2 - 4(9)(2a^2 + 5ab + 2b^2) \\ &= 81(a^2 + 2ab + b^2) - 36(2a^2 + 5ab + 2b^2) \\ &= 81a^2 + 162ab + 81b^2 - 72a^2 - 180ab - 72b^2 \\ &= 9a^2 - 18ab + 9b^2 = 9(a-b)^2 = [3(a-b)]^2 \end{aligned} \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

Since $D = 9(a-b)^2 \geq 0$, the equation has real roots given by $\sqrt{D} = 3(a-b)$ (taking $\pm$ handles sign).

Step 2: Substitute into the quadratic formula:

$$x = \frac{9(a+b) \pm 3(a-b)}{2(9)} = \frac{3(a+b) \pm (a-b)}{6} \quad \textbf{[}\tfrac{1}{2}\textbf{ mark]}$$

Case 1 (+ sign): $x = \dfrac{3a + 3b + a - b}{6} = \dfrac{4a + 2b}{6} = \dfrac{2a + b}{3}$.

Case 2 (− sign): $x = \dfrac{3a + 3b - a + b}{6} = \dfrac{2a + 4b}{6} = \dfrac{a + 2b}{3}$.

Hence, the roots are $\boxed{x = \dfrac{2a+b}{3} \text{ and } x = \dfrac{a+2b}{3}}$. **[1 mark]**

Q8. A cottage industry produces a certain number of pottery articles in a day. It was observed on a particular day that the cost of production of each article (in rupees) was 3 more than twice the number of articles produced on that day. If the total cost of production on that day was Rs. 90, find the number of articles produced and the cost of each article. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let the number of pottery articles produced on that day be x ($x \in \mathbb{N}$).
Then the cost of production of each article = Rs. $(2x+3)$.

Formula Used: Total Cost = (Number of articles) × (Cost per article)

According to the problem:

$$x(2x+3) = 90 \implies 2x^2 + 3x - 90 = 0 \quad \textbf{[1 mark]}$$

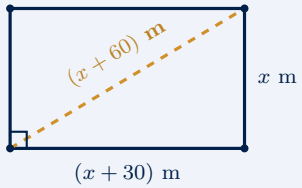
Splitting the middle term ($ac = -180$; $15 \times (-12) = -180$ and $15 - 12 = 3$):

$$2x^2 + 15x - 12x - 90 = 0$$
$$x(2x+15) - 6(2x+15) = 0$$
$$(2x+15)(x-6) = 0 \implies x = -\frac{15}{2} \text{ or } x = 6 \quad \textbf{[1 mark]}$$

Since the number of articles must be a positive integer, we reject $x = -\frac{15}{2}$.
Therefore, number of articles produced = $x = 6$, and cost of each article = $2(6)+3$ = Rs. 15. (Check: $6 \times 15 = 90$.)
Hence, $\boxed{\text{Number of articles} = 6 \text{ and Cost per article} = \text{Rs. } 15}$. **[1 mark]**

Q9. The diagonal of a rectangular field is 60 m more than the shorter side. If the longer side is 30 m more than the shorter side, find the sides of the field. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Let the shorter side of the rectangular field be x m ($x > 0$). Then the longer side is $(x+30)$ m and the diagonal is $(x+60)$ m, forming a right-angled triangle with perpendicular sides x m and $(x+30)$ m, as shown alongside.

Formula Used: Pythagoras' Theorem: (Longer side)^2 + (Shorter side)^2 = (Diagonal)^2

Applying Pythagoras' theorem to the right-angled triangle formed by the sides and diagonal:

$$(x+30)^2 + x^2 = (x+60)^2$$
$$(x^2 + 60x + 900) + x^2 = x^2 + 120x + 3600 \quad \textbf{[1 mark]}$$
$$2x^2 + 60x + 900 - x^2 - 120x - 3600 = 0$$
$$x^2 - 60x - 2700 = 0 \quad [\tfrac{1}{2} \textbf{ mark}]$$

Splitting the middle term ($-90 \times 30 = -2700$ and $-90 + 30 = -60$):

$$x^2 - 90x + 30x - 2700 = 0 \implies x(x-90) + 30(x-90) = 0 \implies (x-90)(x+30) = 0 \quad \textbf{[1 mark]}$$

Thus $x = 90$ or $x = -30$. Since length of a side cannot be negative ($x > 0$), we reject $x = -30$.
Hence, Shorter side = 90 m and Longer side = $90 + 30 = 120$ m (with diagonal 150 m).
Therefore, the sides of the rectangular field are $\boxed{90 \text{ m and } 120 \text{ m}}$. $[\tfrac{1}{2} \textbf{ mark}]$

Q10. Case Study: Graphic Design Studio & Banner Enlargement

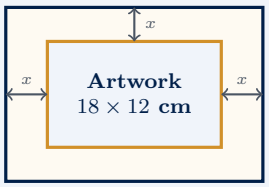
A graphic design studio creates a rectangular poster of dimensions 18 cm × 12 cm. For an exhibition banner, the client requests a uniform margin of width x cm ($x > 0$) on all four sides around the poster so that the total area of the enlarged banner becomes 576 cm^2. [4]

(i) Express the outer length and breadth of the enlarged banner in terms of x. **[1]**

(ii) Form and simplify the quadratic equation in x for the enlarged banner area of 576 cm^2. **[1]**

(iii) Find the uniform margin width x and dimensions of the new banner. **OR** Find the percentage increase in total area of the banner over the original poster. **[2]**

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



New length: $(18 + 2x)$ cm

As shown in the diagram alongside, the inner poster rectangle measures 18 cm × 12 cm (area 216 cm^2). Adding a uniform border of width x cm on all four sides increases each linear dimension by $2x$ cm, resulting in an outer exhibition banner of total area 576 cm^2.

Formula Used: Area = Length × Breadth, $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

(i) New Dimensions: Since a margin of width x cm is added on both opposite sides along length and breadth:

$$\boxed{\text{Outer length} = (18 + 2x) \text{ cm}} \quad \text{and} \quad \boxed{\text{Outer breadth} = (12 + 2x) \text{ cm}} \quad \textbf{[1 mark]}$$

(ii) Quadratic Equation: Given enlarged area = 576 cm^2:

$$(18 + 2x)(12 + 2x) = 576 \implies 4(9 + x)(6 + x) = 576 \implies (x + 9)(x + 6) = 144$$

$$x^2 + 15x + 54 - 144 = 0 \implies \boxed{x^2 + 15x - 90 = 0} \quad \textbf{[1 mark]}$$

(iii) First Part (Margin Width & New Size): Applying the quadratic formula ($a = 1, b = 15, c = -90$):

$$x = \frac{-15 \pm \sqrt{15^2 - 4(1)(-90)}}{2(1)} = \frac{-15 \pm \sqrt{585}}{2} = \frac{-15 \pm 3\sqrt{65}}{2}$$

Since margin width $x > 0$, we reject $\dfrac{-15 - 3\sqrt{65}}{2} < 0$. Thus $x = \dfrac{3\sqrt{65} - 15}{2}$ cm $\approx \dfrac{3(8.0623) - 15}{2} \approx$ 4.59 cm.

New dimensions: Length $= 18 + 2x = (3\sqrt{65} + 3)$ cm ≈ 27.19 cm and Breadth $= 12 + 2x = (3\sqrt{65} - 3)$ cm ≈ 21.19 cm.

Hence, $\boxed{x = \dfrac{3\sqrt{65} - 15}{2} \text{ cm } (\approx 4.59 \text{ cm}),\ \text{New Size } \approx 27.19 \text{ cm} \times 21.19 \text{ cm}}$. **[2 marks]**

OR (Alternative Part — Percentage Increase in Area): Original area = $18 \times 12 = 216$ cm^2; new area = 576 cm^2, so increase = $576 - 216 = 360$ cm^2.

$$\text{Percentage Increase} = \frac{360}{216} \times 100\% = \frac{5}{3} \times 100\% = \frac{500}{3}\% = \boxed{166\frac{2}{3}\% \ (\approx 166.67\%)} \quad \textbf{[2 marks]}$$

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